

# EXPIREMENT - 1 8 BIT ADDITION

PROGRAM:

MNEMONICS

LDA 8050

MOV B, A

LDA 8051

ADD B

STA 8052

HLT

Explanation

Load accumulator with first number in address 8050

Move the data from accumulator to 'B' register

Load accumulator with second number in the address 8051.

Add the data of B register with accumulator

store the data of output in 8052 address.

Halt

AIM:

to write an assembly language program to implement 8-bit addition using 8085 processor.



## ALGORITHM:

1. start the program by loading the first data into accumulator.
2. move the data to register
3. Get the second data and load it into the accumulator
4. add the two register contents.
5. check for carry.
6. store the value of sum and carry in the memory location.
7. HALT.

## INPUT:

| ADDRESS | DATA |
|---------|------|
| 8050    | 1    |
| 8051    | 2    |

## OUTPUT:

| ADDRESS | DATA |
|---------|------|
| 8052    | 3    |

## RESULT:

Thus the program was executed successfully using 8085 processor simulator



